

Urban Crime Insights

A Data-Driven Study of Bicycle Theft Trends and Prevention in Toronto

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Abstract—Toronto continues to face a major problem with bicycle theft. Data from the Toronto Police Service shows that over 2,000 bicycles have been stolen so far this year.[1] Our work aims to identify the key patterns behind bicycle thefts in Toronto by analyzing open data from the Toronto Police Service. By understanding where, when, and why thefts occur, we hope to provide practical insights that support safer cycling and better prevention strategies across the city.

I. INTRODUCTION

A stolen bike is more than lost property, it is lost freedom. This idea apprehends the increasing concern around bicycle theft in urban cities like Toronto. Bicycles are a popular option for people of all ages because they are designed for longer distances, better performance, and less physical strain. However, theft has increased along with the number of bicycles on the road. Bicycles are very hard to find once stolen because they can be easily transported and concealed. Better and more economical ways to safeguard bikes and assist their owners are therefore desperately needed. Current security technologies, such as CCTV cameras, alarms, and secure building entrances, help reduce general crime but often fail when it comes to bicycle theft[2]. These systems are good at recording the theft, however they can rarely prevent it.

II. PROJECT SCOPE

Data-Driven Scope of Bicycle Theft Analysis

Instead of viewing each theft as an isolated event, we target to examine the broader patterns that connect them. The data allows us to see where thefts are occurring most often, such as specific neighborhoods, police divisions, and high-traffic commuter areas. It also shows when thefts are most likely to occur, including seasonal trends, specific days of the week, and certain times of day. We also study how different bicycle characteristics, such as type, brand, cost, and speed are related to theft risk. By exploring these patterns, the project offers insights that can help cyclists choose comparatively safer parking locations and times, assist property managers in improving building and bike-room security, help city planners in designing safer cycling infrastructure, and support policymakers in creating informed strategies to reduce theft. The

hypothesis can also increase public awareness and motivate preventative actions that can lower theft rates. Through this data-driven approach, the project transforms complex information into practical knowledge that can strengthen safety, prevent loss, and safeguard Toronto's growing cycling community.

III. DATASET

The dataset used in this study is a CSV file containing 39,473 bicycle theft records obtained from the Toronto Police Service Public Safety Data Portal. Each record includes detailed temporal information (such as occurrence date, hour, and day of week), spatial attributes (including police division, neighbourhood, and GPS coordinates), and item-specific details about the stolen bicycle such as make, model, type, colour, speed, and estimated cost. Additional context is provided through fields describing the location type, premises type, and case status (e.g., stolen or recovered). This rich collection of structured attributes enables a comprehensive analysis of when, where, and how bicycle thefts occur in Toronto.

IV. DATA PREPROCESSING

A thorough preprocessing and cleaning procedure was applied to ensure the quality and reliability of the bicycle theft dataset before analysis.

Handling Missing Data

Missing or invalid dates in `OCC_DATE` and `REPORT_DATE` were identified and removed, as these fields are critical for temporal analysis. Categorical fields such as `BIKE_MAKE`, `BIKE_MODEL`, `BIKE_COLOUR`, `BIKE_TYPE`, `LOCATION_TYPE`, and `PREMISES_TYPE` contained missing values, which were imputed using the label "UNKNOWN" to preserve all records. Numerical fields (e.g., `BIKE_SPEED` and `BIKE_COST`) were converted to numeric types; zeros were treated as missing values and replaced using median imputation to ensure realistic values. As part of data preprocessing 400 records were dropped.

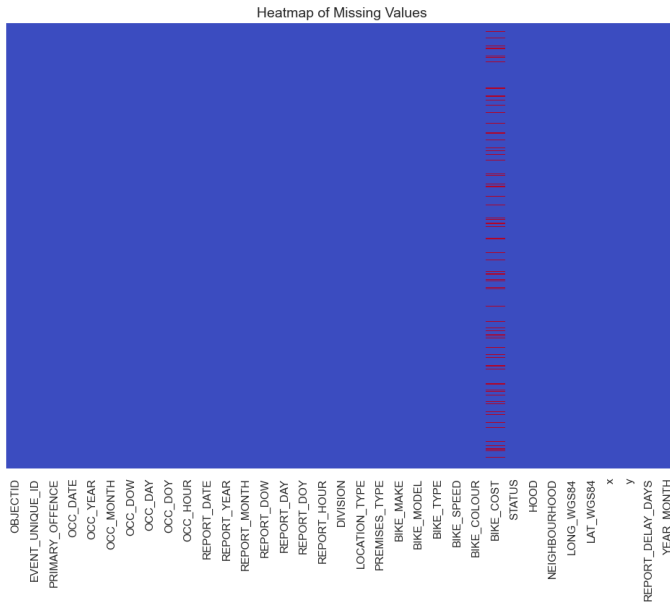


Fig. 1. Heatmap showing missing values across all features in the dataset.

Outliers Detection and Treatment

Cost-related outliers in `BIKE_COST` were detected using the Interquartile Range (IQR) method. The first quartile (Q1) and third quartile (Q3) were used to compute the IQR, and both a lower bound ($Q1 - 1.5 \times IQR$) and an upper bound ($Q3 + 1.5 \times IQR$) were applied. Any bicycle cost falling below the lower bound or above the upper bound was classified as an outlier and removed. This approach eliminates extremely low, unrealistic, or negative cost values as well as abnormally high values, ensuring a more reliable distribution of bicycle cost data for analysis.

Data Transformation

All date fields were converted into standardized datetime formats to support accurate time-based operations. New temporal features such as `YEAR_MONTH` and `REPORT_DELAY_DAYS` were engineered to enable trend visualization and reporting delay analysis. Numeric fields were cleaned and normalized through imputation to maintain consistency across the dataset.

Data Cleaning and Feature Engineering

The preprocessing pipeline included removing duplicate rows, dropping historical records prior to 2014, and eliminating redundant columns (e.g., outdated neighbourhood variables). Invalid spatial records (where LAT_WGS84 or LONG_WGS84 were equal to 0) were removed to ensure accurate mapping. Feature engineering introduced useful variables—such as reporting delay and monthly time periods—to enhance analytical insights.

Documentation of Preprocessing Steps

Each preprocessing decision—handling missing values, applying IQR-based outlier removal, engineering features, and cleaning structural issues—was clearly logged and justified. Proper documentation ensures the workflow is transparent, reproducible, and easy for others to follow in future analyses.

Dataset Overview-EDA

To obtain a high-level understanding of the data, an overview function was used to report the total number of rows and columns, the number of duplicate records, and the count of missing values per feature. This helped validate the effectiveness of the preprocessing pipeline and confirmed that the cleaned dataset had a consistent structure with relatively few remaining missing values.

Numerical Summary

Missing or invalid dates in `OCC_DATE` and `REPORT_DATE` were identified and removed, as these fields are critical for temporal analysis. Categorical fields such as `BIKE_MAKE`, `BIKE_MODEL`, `BIKE_COLOUR`, `BIKE_TYPE`, `LOCATION_TYPE`, and `PREMISES_TYPE` contained missing values. Traditional imputation techniques such as mean, median, and mode are not well suited for categorical data; mean and median are not applicable, and using mode can artificially inflate the most frequent category, leading to inaccurate patterns. To avoid this distortion, all missing categorical values were replaced with the label “UNKNOWN,” ensuring that no artificial bias was introduced into category distributions.

Numerical fields (e.g., `BIKE_SPEED` and `BIKE_COST`) were converted to numeric types; zeros were treated as missing values and replaced using median imputation to maintain realistic and consistent values without being overly influenced by extreme outliers.

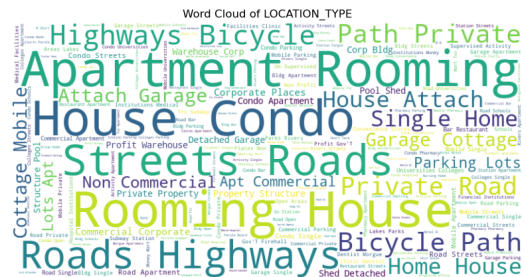


Fig. 2. Word cloud of LOCATION_TYPE, illustrating the most frequent locations associated with bicycle thefts. Larger words indicate locations reported more often.

Temporal Patterns and Trends 1

Temporal patterns were examined using the occurrence year (OCC_YEAR), month (OCC_MONTH), day of week (OCC_DOW), and hour of day (OCC_HOUR). The most common (mode) values for each of these fields were used to

identify peak theft periods in terms of seasonality, weekday versus weekend, and time of day. In addition, the engineered `YEAR_MONTH` feature allows monthly aggregation for trend analysis over time, making it easier to observe any long-term increases, decreases, or seasonal cycles in bicycle thefts.

Relationship Between Numerical Features

The correlation heatmap shows that none of the numerical features have strong correlations with each other, indicating that these variables are largely independent.



Fig. 3. Correlation heatmap illustrating the relationships between numerical variables in the dataset.

Categorical Feature Association (Cramér's V)

Cramér's V shows that most categorical features have very weak relationships with each other, meaning the categories do not strongly influence one another.

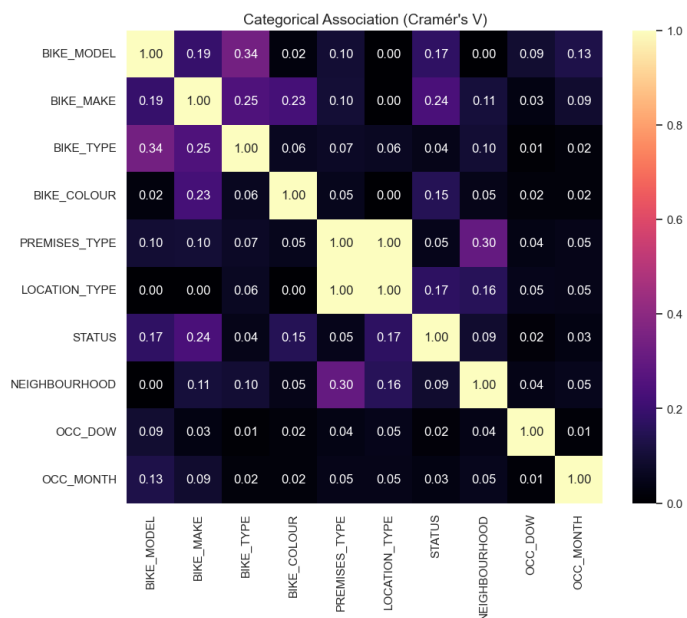


Fig. 4. Cramér's V association heatmap for categorical variables in the dataset.

Pairwise Relationships Between Numerical Features

This pair plot shows how each numerical feature changes with the others. The points are spread out randomly, which means there are no strong patterns or relationships between these variables.

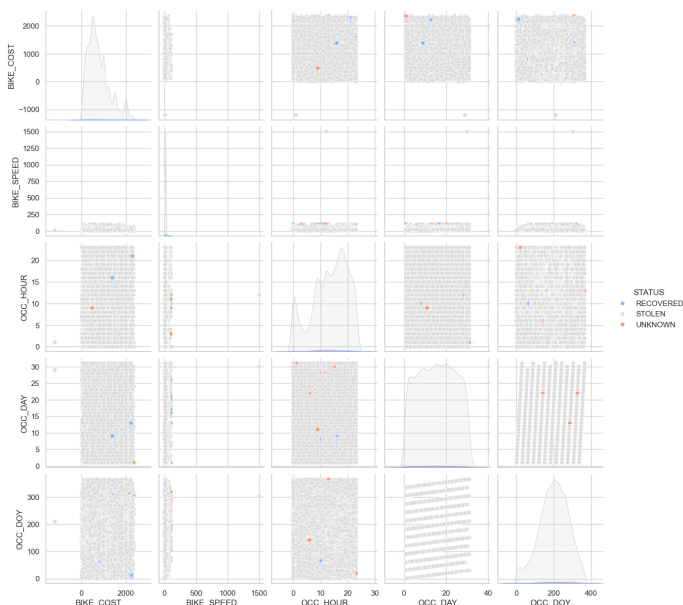


Fig. 5. Pair plot showing distributions and pairwise relationships among numerical variables.

Temporal Patterns and Trends 2

This plot shows that bicycle thefts occur most often in the late afternoon and evening, especially between 3 PM and 8

PM. Theft activity is lowest during the early morning hours.

Key Findings from EDA

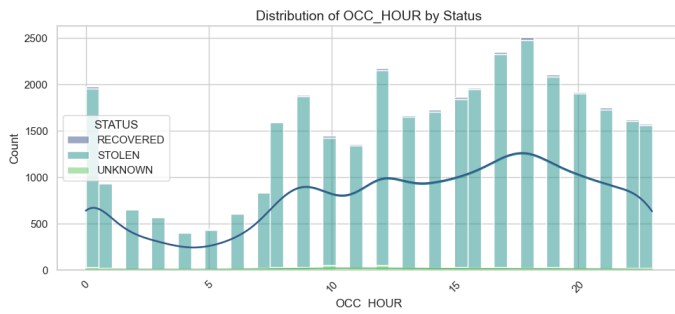


Fig. 6. Distribution of OCC_HOUR by theft status.

Temporal Patterns and Trends 3

This plot shows that bicycle thefts peak in the summer months—especially June to August—and drop sharply during the winter.

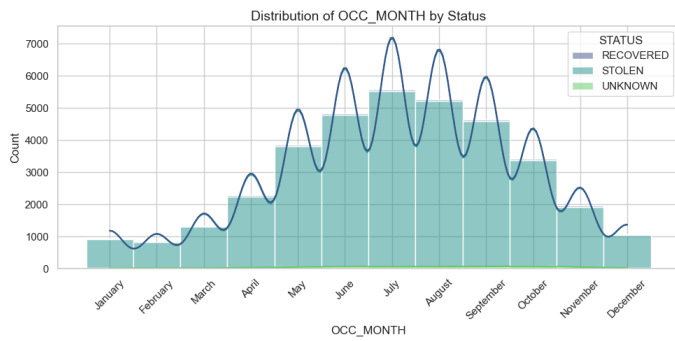


Fig. 7. Distribution of OCC_MONTH by theft status.

Distribution of Bicycle Cost

This plot shows that most stolen bicycles are in the mid-cost range, while very low-cost and high-cost bikes are stolen less often.

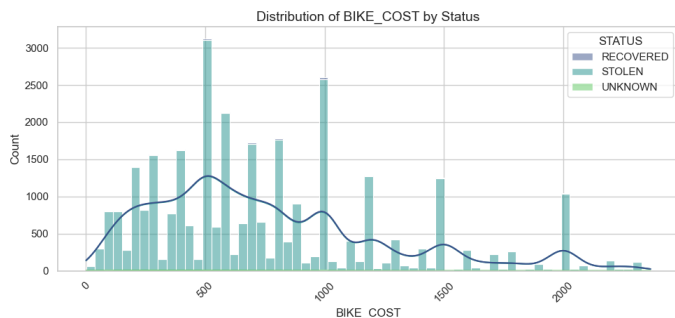


Fig. 8. Distribution of BIKE_COST by theft status.

The dataset initially contained 39,473 bicycle theft records with 35 attributes. Key preprocessing steps included converting date fields to proper datetime formats, removing 59 historical entries and 341 invalid spatial records, and eliminating redundant neighbourhood columns. Missing categorical values were replaced with “Unknown”, while numerical issues in BIKE_SPEED and BIKE_COST were addressed using median imputation and IQR-based outlier removal. Feature engineering introduced REPORT_DELAY_DAYS and YEAR_MONTH to support temporal trend analysis. After cleaning, the dataset contained 39,073 valid entries with improved consistency.

Exploratory analysis revealed that the most frequently stolen bike makes were OT, UK, GI, TR, and OTHER, while the most common bike types included MT (Mountain), RG (Road/General), OT, RC, and EL (Electric). These patterns show that a wide range of bicycle categories are targeted. Overall findings highlighted a very low recovery rate (1%), clear seasonal theft peaks in July, and high-risk neighbourhoods such as the Yonge–Bay Corridor.

From the exploratory analysis, several key insights emerged. First, the recovery rate of stolen bicycles, computed using the STATUS field, was found to be very low, indicating that the vast majority of bikes are never recovered once stolen. Second, the temporal analysis revealed clear peak periods, with thefts clustering around specific days of the week and hours of the day, as well as particular months of the year. Third, certain neighbourhoods appeared much more frequently than others, suggesting spatial hotspots for bicycle theft. Together, these findings provide an evidence-based foundation for subsequent visualizations and can help inform targeted prevention strategies.

- 1) **Recovery Rate:** Only 1% of stolen bicycles are ever recovered, indicating extremely low retrieval success.
- 2) **Peak Theft Time:** The highest concentration of thefts occurs on Fridays around 18:00, with additional increases during late-afternoon hours.
- 3) **Seasonal Peak:** July represents the peak month for bicycle thefts, reflecting significantly higher activity during the summer season.
- 4) **Peak Year:** Theft activity reached its highest level in 2018.
- 5) **High-Risk Neighbourhoods:** The Yonge–Bay Corridor is the most affected area, followed by Downtown Yonge East, St. Lawrence–Bayfront–The Islands, Wellington Place, and the Annex.
- 6) **Most Targeted Bicycle Types:** Black, mountain, and road bicycles are stolen most frequently, and the most commonly stolen makes include UK and OT.
- 7) **High-Risk Premise Settings:** Outdoor areas and apartment or condo premises present elevated theft risk, particularly during summer months, weekdays, and late-afternoon periods.

Multivariate 3D Scatter Analysis: BIKE_COST, BIKE_SPEED, and OCC_HOUR

The 3D scatter plot reveals that the majority of stolen bicycles cost approximately \$1000 or less and have speeds below 40 km/h. These incidents occur more frequently during the later hours of the day, indicating increased vulnerability during evening periods. Overall, the visualization suggests that moderately priced, lower-speed bicycles are the most common theft targets, with little distinction across recovery statuses. The concentration of bicycle thefts at specific times of day suggests a pattern more aligned with *professional or planned theft* rather than random opportunistic incidents. This indicates that some offenders may be deliberately targeting predictable time windows when bicycles are most vulnerable.

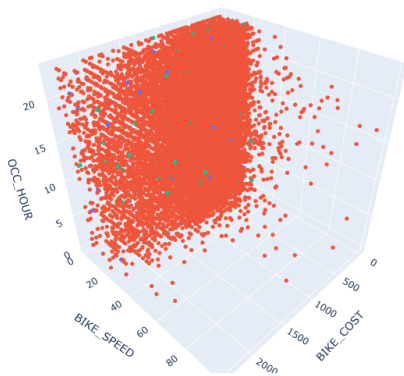


Fig. 9. 3D scatter plot showing the relationship between bike cost, bike speed, and time of theft. Most stolen bikes are under \$1000, have speeds below 40 km/h, and are taken later in the day.

Neighbourhood-Based Spatial Clustering of Bicycle Thefts

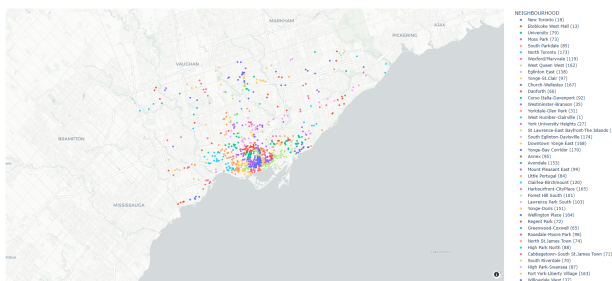


Fig. 10. Geospatial Clustering of Bicycle Thefts Across Toronto Neighbourhoods

The neighbourhood cluster map shows clear spatial patterns in bicycle theft occurrences across Toronto. Most thefts are heavily concentrated in the downtown core, particularly around high-density and high-traffic neighbourhoods such as Downtown Yonge, Waterfront, Trinity-Bellwoods, and Church-Wellesley. Suburban regions including Etobicoke, North York,

and Scarborough exhibit significantly lower theft counts, suggesting reduced crime density outside the city centre. The clustering aligns with major transit corridors, commercial districts, and university zones, indicating that mobility and population density influence theft risk. These spatial insights help identify neighbourhoods that may benefit from targeted prevention strategies such as secure bike infrastructure or increased surveillance.

Recovery Patterns by Reporting Delay

Bikes are most likely to be recovered when the theft is reported on the same day. The longer someone waits to report the theft, the lower the recovery rate becomes. Reporting delays of more than a week lead to very low chances of getting the bike back.

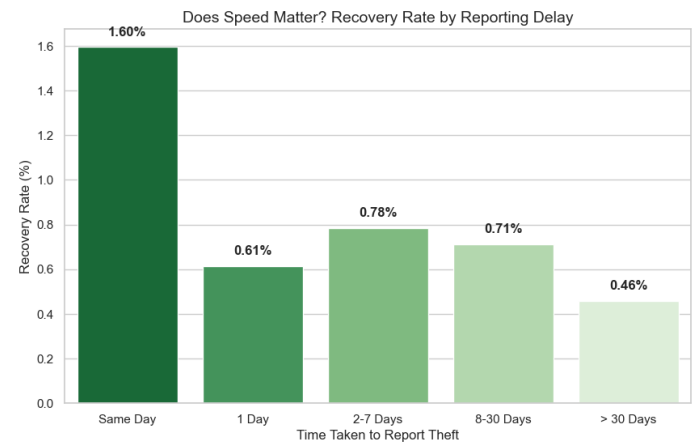


Fig. 11. Recovery rate based on how long it takes to report the bicycle theft.

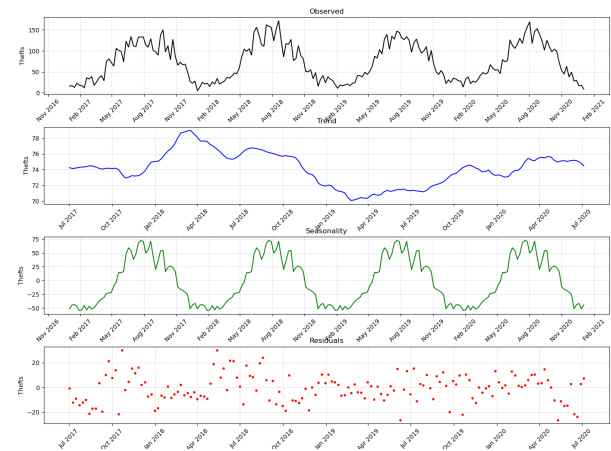


Fig. 12. Time-series decomposition of bicycle thefts, highlighting the long-term trend, strong summer seasonality, and residual random fluctuations.

V. EVALUATION METRICS AND MODEL INTERPRETATION

Evaluating the machine learning models using accuracy, precision, recall, F1-score, and ROC-AUC provides insight

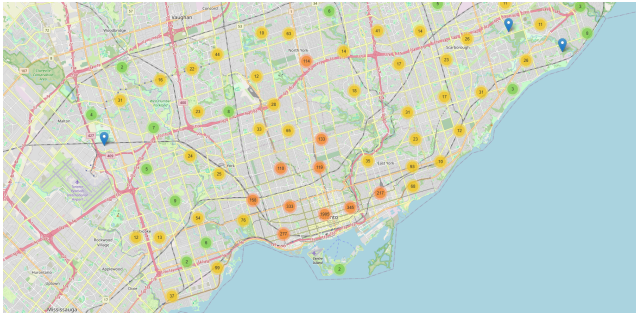


Fig. 13. Cluster map showing geographic hotspots of bicycle thefts across Toronto. Larger and darker clusters indicate areas with higher concentrations of theft incidents.

into how each algorithm behaves under the extreme class imbalance (approximately 99% non-recovered vs. 1% recovered bicycles). These metrics help explain why certain models perform poorly and emphasize the need for additional preprocessing.

Logistic Regression:

- **Accuracy:** 0.59 — very low; the model is unstable due to the severe imbalance.
- **Precision:** 0.013 — almost every predicted positive is incorrect.
- **Recall:** 0.504 — detects some positive cases but inconsistently.
- **F1-score:** 0.024 — extremely poor harmonic performance.
- **ROC AUC:** 0.57 — slightly above random.

Conclusion: Logistic Regression struggles significantly, showing high confusion and strong sensitivity to skewed data.

Random Forest:

- **Accuracy:** 0.99 — appears high but is misleading due to the dominance of class 0.
- **Precision:** 0.556 — roughly half of its predicted positives are correct.
- **Recall:** 0.128 — identifies only 12.8% of actual positives.
- **F1-score:** 0.208 — weak overall performance.
- **ROC AUC:** 0.611 — slightly above random classification.

Conclusion: Random Forest performs slightly better but remains strongly biased toward the majority class.

Gradient Boosting:

- **Accuracy:** 0.989 — high but not meaningful under class imbalance.
- **Precision, Recall, F1-score:** 0.0 — the model never predicts the positive class.
- **ROC AUC:** 0.636 — shows some separation ability but does not translate into positive predictions.

Conclusion: Gradient Boosting entirely ignores the minority class and fails to identify any positive cases.

Overall Interpretation

All three models struggle to correctly classify the minority class because the dataset is extremely imbalanced, with only

about 1% positive cases. Accuracy becomes misleading under these conditions, while precision, recall, and F1-score clearly show the models' inability to detect rare events. Although Random Forest performs marginally better, it still captures only a small fraction of positive cases.

These results indicate that meaningful predictive modeling is not possible without first addressing the class imbalance.

Using Better Evaluation Metrics

Since accuracy is misleading in highly imbalanced datasets, it is important to rely on metrics that better reflect performance on rare classes. Useful alternatives include:

- **Balanced Accuracy** — averages sensitivity and specificity to correct for skewed class distribution.
- **Confusion Matrix** — provides a complete view of true/false predictions for both classes.

These evaluation metrics offer a clearer picture of model effectiveness for rare events such as bicycle recovery.

VI. SOUNDNESS OF RESULTS AND COMPARISONS

To ensure the reliability and scientific soundness of the findings, multiple verification steps were applied:

- **Cross-Checking Across Visualizations:** Key patterns were confirmed using different charts. For example, both the EDA and the seasonal decomposition showed a peak in bicycle thefts during July, reinforcing the conclusion of strong summer seasonality. This is logically consistent, as warmer months lead to more outdoor riding and storage, increasing theft opportunities.
- **Multi-Visualization Consistency:** Patterns such as neighbourhood hotspots, seasonal trends, and common bike characteristics were validated across heatmaps, trend lines, spatial maps, and distribution plots. This avoided reliance on any single visualization.
- **Context-Aware Interpretation:** All insights were evaluated considering dataset limitations and domain context, ensuring that conclusions remained realistic and logically grounded.
- **Logical Data Cleaning:** Removing impossible values (such as negative bike costs or spatial coordinates recorded as (0,0)) preserved the mathematical and interpretive integrity of all analyses, including means, medians, correlations, and spatial distance patterns.
- **Overall Soundness:** The combined use of cross-validation, comparative analysis, and rigorous data cleaning ensures that the findings are consistent, logical, and supported by multiple forms of evidence rather than a single viewpoint.
- **Comparative Feature Analysis:** A comparison was performed to determine which features influence the theft outcome (STATUS) more strongly. Cramér's V revealed that LOCATION_TYPE has an association of 0.17 with STATUS, while BIKE_MAKE has a slightly stronger association of 0.24. This suggests that bike features—often linked to value or theft desirability—are more dominant

explanatory variables than environments under which bike thefts occur.

VII. CONCLUSION

Cyclists should use secure, monitored parking areas and stronger locking mechanisms, especially during evening hours—to reduce theft risk for mid-cost, frequently targeted bicycles. The neighbourhoods of Yonge–Bay Corridor, Downtown Yonge East, St. Lawrence–Bayfront–The Islands, Wellington Place, and the Annex consistently record the highest number of bicycle thefts. These areas are dense, high-traffic zones in Toronto’s downtown core, where increased pedestrian movement, transit access, and mixed residential–commercial activity contribute to elevated theft risk. Cyclists may reduce theft risk by avoiding highly targeted bicycle types, such as black, mountain, or road bikes, when possible or by enhancing their security measures. They should also avoid leaving bicycles in vulnerable premise settings—such as outside or in apartment areas—especially during summer months, on weekdays, and in the late afternoon when theft activity is highest.

Please note that the visualizations discussed in this report are not exhaustive; many additional plots were generated during the analysis and can be explored through the interactive GUI developed for this project.

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- [1] Toronto Police Service, *Bicycle Theft (Public Safety Data Portal)*. Accessed: Dec. 7, 2025. [Online]. Available: <https://data.torontopolice.on.ca/pages/c78364ab031747359fa8afb78febdd3d>.
- [2] T. S. Ripa, T. F. H. Khan, M. Rokunuzzaman, M. R. Rawshan, S. A. Oishi, and K. M. Zim, “Bicycle Theft Detection by Using YOLOv5s and DeepSORT Algorithm,” in *Proc. 2024 Int. Conf. Recent Progresses Sci., Eng. Technol. (ICRPSET)*, Rajshahi, Bangladesh, 2024, pp. 1–5, doi: 10.1109/ICRPSET64863.2024.10955901.